



**Verified Carbon
Standard**

QINGDAO LONGXIN WIND POWER PROJECT PHASE I

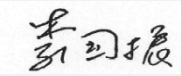


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Report Title	VCS Validation Report of Qingdao Longxin Wind Power Project Phase I
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Summary:

China Quality Certification Center (hereafter referred to as CQC) has been commissioned by Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd. to perform a validation of the renewal of crediting period of “Qingdao Longxin Wind Power Project Phase I” (Project. ID. 969) (hereinafter referred to as the project) in P. R. China.

The assessment was performed in accordance with the VCS standards version 4.3 and UNFCCC requirement including an assessment of:

- (a) The impact of new relevant national and/or sectoral policies and circumstances on the baseline taking into account relevant guidance from the Board with regard to renewal of the crediting period of the registered VCS project activity at the time of requesting renewal of crediting period of the project activity;
- (b) The correctness of the application of the approved methodology and, where applicable, the approved standardized baseline for the determination of the continued validity of the baseline or its update, and the estimation of emission reductions for the applicable crediting period of the registered VCS project activity.

The project has been registered as VCS project with project ID: 969 and VCUs of 82,302 tCO₂e have been issued for the monitoring period from 08/11/2011- 27/09/2012. The Project has been registered as a CDM project on 28/09/2012¹ (UNFCCC registration reference number: 6697). The emission reductions of the project have not been issued as CERs.

Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the Project) developed by China Resources Wind Power (Qingdao) Co., Ltd., involves construction and

¹ <http://cdm.unfccc.int/Projects/DB/TUEV-RHEIN1342414627.77/view>

operation of a wind power project that is sited on Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The project installs 24 wind turbines with capacity of 2000kW each and 1 wind turbine with capacity of 1800kW, providing a total installed capacity of 49.8MW to generate clean renewable electricity.

The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing electricity generated by NCPG which is dominated by fossil fuel fired power plants. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

During this validation, no finding was identified in relation to relevant VCS standards, program guide and UNFCCC requirements and relevant host party criteria and the applied baseline and monitoring methodology etc.

In summary, it is CQC's opinion that the project activity "Qingdao Longxin Wind Power Project Phase I" (Ref. No. 969) in P. R. China, as described in the PD, version 02 dated 29/08/2022, meets all relevant VCS and UNFCCC requirements for the renewal of the crediting period.

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1 INTRODUCTION

1.1 Objective

China Quality Certification Center (hereafter referred to as CQC) has been commissioned by Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd. to perform a validation of the renewal of crediting period of “ Qingdao Longxin Wind Power Project Phase I” (project ID. No. 969) (hereinafter referred to as the Project) in P. R. China.

The validation objective is an independent assessment by a third party of a proposed project activity against all defined criteria set for the renewal of crediting period under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of project conformance to VCS rules, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology, and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCUs. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for renewal of crediting period.

1.2 Scope and Criteria

CQC has employed a risk-based approach in the validation, focusing on the identification of significant risks and reliability of project design and generation of emission reductions according to the relevant applicable rules for validation under the VCS.

A risk-based approach of document review and an on-site visit combined method were followed by CQC to perform this validation.

The validation scope is defined as an independent and objective review of the project description (PD) to validate that: (a) the project design is complete and understandable, (b) environmental and socio-economic impacts are positive, (c) the engagement of stakeholders is sufficient, (d) the baseline scenario is correctly defined as per the applied methodology and relate tools, (e) the project is additional, (f) the monitoring plan can be implemented and is transparent and adequate and (g) all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature.

The project activity, as documented in PD (version 02) is reviewed against the criteria stated in VCS standard Version 4.3 and the approved baseline and monitoring methodology ACM0002 (version 20.0). The validation was based on the requirements

VCS Validation and Verification Manual and VCS programs guide and VCS standard Version 4.3 applying auditing techniques.

The validation is not meant to provide any consulting towards the PP. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

1.3 Level of Assurance

The validation report is based on the VCS-PD and supporting evidences made available to the validator and information collected through performing interviews and during the on-site assessment.

The validation conclusion is assured a reasonable level of assurance.

1.4 Summary Description of the Project

Project title	Qingdao Longxin Wind Power Project Phase I
UNFCCC reference number	6697
VCS project ID	969
Project Participants	China Resources Wind Power (Qingdao) Co., Ltd.
Location of the project	Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. Geographic coordinates: East longitude: from 119.8827° to 119.9503° North latitude: from 36.8157° to 37.8915°
Project start date	Construction date: 26/05/2011; Operation start date: 08/11/2011
First Project Description (PD)	Version 01, dated 09/08/2022
Final Project Description (PD)	Version 02, dated 29/08/2022
Applied Methodology/Version	ACM0002, version 20.0

Scope/Technical Area	1/1.2
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The Project has been registered as VCS project with Ref. VCS 969 and VCU of 82,302 tCO₂e have been issued for the monitoring period from 08/11/2011- 27/09/2012. The project has been registered as a CDM project on 28/09/2012 (UNFCCC registration reference number: 6697). The emission reductions of the project have not been issued as CERs.

Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the Project) developed by China Resources Wind Power (Qingdao) Co., Ltd., involves construction and operation of a wind power project that is sited on Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The Project installs 24 wind turbines with capacity of 2000kW each and 1 wind turbine with capacity of 1800kW, providing a total installed capacity of 49.8MW to generate clean renewable electricity. The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing electricity generated by NCPG which is dominated by fossil fuel fired power plants. The estimated annual net electricity generation and average annual emission reductions of the proposed project are 99,026MWh and 81,885tCO₂e, respectively.

2 VALIDATION PROCESS

2.1 Method and Criteria

The validation method of the project consisted of the following steps:

- Contract review
- Appointment of validation team and technical reviewers
- A desk review of the VCS PD submitted by the client and additional supporting documents with the use of customized validation protocol
- Validation planning
- On-Site assessment
- Background investigation and follow-up interviews with personnel of the project developer and its contractors
- Draft validation reporting
- Resolution of corrective actions and clarifications
- Final validation reporting

- Technical review
- Final approval of the validation.

The validation criteria consisted of the following:

- VCS Standard, v4.3^{/8/}
- VCS Program Guide, v4.2^{/9/}
- Program Definitions, v4.2^{/10/}
- VCS Validation Verification Manual v3.2^{/11/}
- The approved methodology applied in the proposed project activity is ACM0002 (version 20.0) – “Grid-connected electricity generation from renewable sources”^{/12/}
- Tool for the demonstration and assessment of additionality (version 07.0.0)^{/14/}
- Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period”(version 03.0.1)^{/16/}
- Tool to calculate the emission factor for an electricity system (version 07.0)^{/13/}

2.2 Document Review

PP submitted PD (version 01)^{/1/} and supporting background documents related to the project design to CQC on 09-08-2022.

PP submitted PD (version 02)^{/1/} to CQC on 15/09/2022.

Furthermore, the validation team used additional documentation by third parties like host party legislation, technical reports referring to the project design or to the basic conditions and technical data.

The references used in the course of this validation are summarized in Appendix 1.

2.3 Interviews

The objective of the interview process was to solicit important information from personnel related to project and relevant to the validation process. On-site interviews and information discussions were conducted with PP (China Resources Wind Power (Qingdao) Co., Ltd.), local residents and VCS project development consultant. The interviews were performed by the validation team on-site and the following Table 2-1 is a list of the main interviewees and subject.

Table 2-1: Interview and Interview Topics

Interviewed personnel	Role	Organization	Subject
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Mr. Ju Bin	Project Manager	China Resources Wind Power (Qingdao) Co., Ltd.	Basic information, technology of the project, etc.;
Mr. Shang Zitong	Staff	China Resources Wind Power (Qingdao) Co., Ltd.	Monitor Data: meter readings, control and maintenance, QA&QC systems
Mr. Pang Jianjian	Staff	China Resources Wind Power (Qingdao) Co., Ltd.	Status of the project activity and any modifications with respect to the registered PD.
Mr. Zhao Jie	Staff	China Resources Wind Power (Qingdao) Co., Ltd.	Applicability to the latest methodology.
Mr. Men Ruochen	Project Manager	Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.	National and local policies and changes Baseline of the project and its updates
Ms. Ding Xue	Technical director	Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.	The lifetime of the project activity Emission Factors and their updates Monitoring plan and changes.

2.4 Site Inspections

The assessment team performed the on-site validation (Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P. R. China.) on 09/08/2022. The interviewed personnel and objective are listed in above table.

2.5 Resolution of Findings

Material discrepancies identified in the course of the validation are addressed either as CARs, CLs or FARs.

A Corrective Action Request (CAR) is established where:

- mistakes have been made in assumptions, application of the methodology or the project documentation which will have a direct influence the project results,

- the requirements deemed relevant for validation of the project with certain characteristics have not been met or,
- there is a risk that the project would not be registered or that emission reductions would not be able to be verified and certified.

A Clarification Request (CL) will be issued where information is insufficient, unclear or not transparent enough to establish whether a requirement is met.

A Forward Action Request (FAR) will be issued when certain issues related to project implementation should be reviewed during the first verification.

There are no CARs and CLs raised for the project.

2.5.1 Forward Action Requests

As this is the validation for the project for renewal of crediting period, there are no FARs raised in the previous validation and verification process.

3 VALIDATION FINDINGS

3.1 Project Details

The following description of the project as per PD was verified during the on-site visit:

Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the Project) developed by China Resources Wind Power (Qingdao) Co., Ltd., involves construction and operation of a wind power project that is sited on Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The geographical coordinates of the project is from 119.8827°E to 119.9503°E, and from 36.8157°N to 37.8915°N (The central GPS coordinate is 119.91125°E, 36.8568°N). The Project installs 24 wind turbines with capacity of 2000kW each and 1 wind turbine with capacity of 1800kW, providing a total installed capacity of 49.8MW to generate clean renewable electricity.

The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing electricity generated by NCPG which is dominated by fossil fuel fired power plants. The estimated annual net electricity generation and average annual emission reductions of the proposed project are 99,026MWh and 81,885tCO₂e respectively in the second crediting period.

The main technical specifications of the wind turbine are provided in the following table.

Item	Unit	Index	Index
Model	-	V90-2.0MW	V90-1.8MW

Rated capacity	kW	2000	1800
Number of blades	-	3	3
Rotor diameter	M	90	90
Rated wind speed	m/s	13	12
Shutoff speed	m/s	2.5	3.5
Height of hub	m	80	80
Rated	V	690	690

By checking the registered CDM PD and site visit interview with project owner, it is confirmed that the project started operation on 08/11/2011 which is the project start date according to the requirement of VCS. Also, the start date of the first VCS crediting period of the project has been set started from 08/11/2011 to 07/11/2021 lasting for 10 years in line with the requirement of VCS.

In the registered VCS PD, the crediting period is described as from 08/11/2011 to 27/09/2012. A deviation is requested for the crediting period in the registered VCS PD. The project was registered under VCS Standard Version 3 and completed validation before 19/03/2020. Thus, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project was updated from 08/11/2011 - 27/09/2012 to 08/11/2011-07/11/2021.

Before the implementation of the project activity, the electricity generated by the project would be supplied by NCPG in the baseline scenario.

By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

The project activity was registered as a CDM project activity on 28/09/2012. All above information is available at <http://cdm.unfccc.int/Projects/DB/TUEVRHEIN1342414627.77/view>.

The project does not participate in the other emissions trading program by checking public information on Internet and interviewing with project owner. Thus, there is no rejection from other GHG programs.

As the project is not a grouped project, the eligibility criteria for grouped project is not applicable; No leakage management as not an AFOLU project; No commercially sensitive information confirmed through site visit.

The project activity would contribute sustainable development in the region in following aspects confirmed by site visit:

The project contributes SDGs Target 7 “Affordable and Clean Energy” by the utilization of wind power as a renewable energy source. The project creates direct and indirect employment opportunities during construction and operation phases, so it contributes to SDGs Target 8 “Decent Work and Economic Growth”. The project produces clean renewable energy by diminishing CO₂ emissions. Therefore, it contributes to SDGs Target 13 “Climate Action”.

Overall, it is confirmed that the PD is accurate, complete, and provides an understanding of the nature of the project.

3.2 Safeguards

3.2.1 No Net Harm

The environmental impact assessment for this project was completed in September 2010 and approved by Qingdao Environmental Protection Bureau on 12/10/2010. The approval number is Qinghuanshen [2010] 209.

According to EIA and the approval, no significant environmental impacts are discovered. With mitigation controls listed as above, the impacts on environment during the construction and operation period will be minimized to meet the requirements of the laws and regulations. As the project is a wind power plant, the project will generate electricity from renewable sources which can reduce the consumption of fossil fuel and GHG emission. Furthermore, the project provides long-term opportunities for local residents and all the employees will be regularly trained with the wind farm operation technology, which contributes to local economic and social sustainable development.

3.2.2 Local Stakeholder Consultation

The local stakeholder’s meeting was held in project site on 20th March, 2011. 50 participants respectively from the local government, surrounding villages attended the meeting. The project owner introduced the proposed project, and then a survey was arranged through a one-page questionnaire, which was designed to be easily filled in. The opinions expressed by the stakeholders were recorded and are available on request.

The stakeholder meeting and the survey showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction. Communications with Local stakeholders was being carried out at periodic intervals. There are no negative comments received for the project.

All such conclusion has been verified through site visit and check registered PD.

3.2.3 Environmental Impact

According to EIA and the approval, no significant environmental impacts are discovered. As the project is a wind power plant, the project will generate electricity from renewable sources which can reduce the consumption of fossil fuel and GHG emission. Furthermore, the project provides long-term opportunities for local residents and all the employees will be regularly trained with the wind farm operation technology, which contributes to local economic and social sustainable development.

3.2.4 Public Comments

There are no public comments received.

3.2.5 AFOLU-Specific Safeguards

Not applicable as non-AFOLU project.

3.3 Application of Methodology

3.3.1 Title and Reference

The approved consolidated baseline and monitoring methodology ACM0002: "Consolidated methodology for grid-connected electricity generation from renewable sources" (Version 20.0)

Related tools are:

Tool for the demonstration and assessment of additionality (version 07.0.0)

Tool to calculate the emission factor for an electricity system (version 07.0)

Tool to determine the remaining lifetime of equipment (Version 01)

Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period (version 03.0.1)

For detailed information please refer to: <https://cdm.unfccc.int/methodologies/index.html>.

3.3.2 Applicability

The criteria and assessment of ACM0002 (version 20.0) are in the following table:

Criteria	Analysis	Assessment team's opinion
This methodology is applicable to grid-connected renewable energy power	The project is the installation of a wind	By site visit and checking FSR, it is

generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	power project. Therefore, (a) is applicable.	confirmed that the project is a Greenfield power plant.
The methodology is applicable under the following conditions: (a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The project activity includes a wind power plant. Therefore, (a) is applicable.	By site visit and checking FSR, it is confirmed that the project is a Greenfield power plant.
In case of hydro power plants, one of the following conditions shall apply:	Not applicable. the project is not a hydro power	By site visit and checking FSR, it is

(a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	project.	confirmed that the project is not a hydro power project, this is not applicable.
In the case of integrated hydro power	Not applicable. the project	By site visit and

projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	is not an integrated hydro power project.	checking FSR, it is confirmed that the project is not an integrated hydro power project, this is not applicable.
The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project activity does not involve switching from fossil fuel to renewable energy sources at the site of the project activity. The project is not a biomass fired power plant/unit.	By site visit and checking FSR, it is confirmed that the project is a Greenfield power plant, does not involved in switching from fossil fuel to renewable energy and biomass fired power plant.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of	Not applicable. the project is not retrofit, rehabilitation, or capacity	By site visit and checking FSR, it is confirmed that the project is a Greenfield

the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	addition project.	power plant.
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According to the table above, the assessment team confirm the project meets all the applicable criteria of ACM0002 version 20.0.

Also, for related tools, the assessment team confirm all applicable criteria have been met, which is confirmed by site visit and checking FSR.

3.3.3 Project Boundary

Project boundary has been ascertained and confirmed during the site visit using ACM0002 version 20.0 – “The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to”.

Hence the project boundary includes the renewable energy power generation, sub-stations, grid and all power plants connected to grid. The proposed project activity will evacuate power to the grid. The boundary also extends to the project power plant and all power plants connected physically to the electricity system that the VCS project power plant is connected to.

The calculation of net electricity supplied to grid is under purview of state electricity board and project activity Owner or project activity Implementer does not have any control on it. Thus, for project activity, net electricity supplied to grid is the monitoring parameter which is used for ER calculations.

The sources and GHG gases involved for proposed Project activity are as below

Source		Gas	Included?	Justification/ Explanation
Baseline	Grid Connected electricity generation.	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
		Other	No	No other emissions are emitted from the project
Project	Greenfield Hydropower Project Activity.	CO ₂	No	No CO ₂ emissions are emitted from the project
		CH ₄	No	Project activity does not emit CH ₄

		N ₂ O	No	Project activity does not emit N ₂ O
		Other	No	Project activity does not emit other forms of GHG emissions

3.3.4 Baseline Scenario

The baseline scenario of the project activity is that the same electricity output by the project activity would have otherwise been generated by the operation of NCPG connected power plants and by the addition of new generation sources. According to the methodology ACM0002 version 20.0, the baseline emissions are the electricity produced by the project activity multiplied by the emission factor of NCPG.

The information presented in the updated PD has been validated by an initial document review of all data. Further confirmation has been made based on the review of information from similar projects and/or technologies. The sources referenced in the PD have been quoted correctly.

The steps from the Methodological Tool “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” version 03.0.1 as per VCS standard version 4.3 were applied to assess the continued validity of the baseline and/or to update the baseline at the renewal of a crediting period:

Step 1: Assess the validity of the current baseline for the next crediting period

VCS standard version 4.3 requires assessing the impact of new relevant national and/or sectoral policies and circumstances on the baseline. The validity of the current baseline is assessed using the following Sub-steps:

Step 1.1: Assess compliance of the current baseline with relevant mandatory national and/or sectoral policies

In China, the Renewable Energy Law has been put into effect since 2006, which encourages the development of renewable energy projects. However, although renewable energy projects have been developed rapidly in recently years, grid connected power generation in China is still dominated by fossil-fuel power plants. There are no new relevant national and/or sectoral policies and/or circumstances in the electricity generation sector applicable to the project activity, in comparison to the time of the submission of the project activity for validation, which would affect the compliance of the current baseline scenario. Hence in the absence of the project activity electricity would still have been generated in the existing fossil fuel power plants or by the addition of new fossil fuel power plants connected to the NCPG.

CQC confirms that no relevant mandatory national and/or sectoral policies applicable to the project activity came into effect after the submission of the project activity for validation.

Step 1.2: Assess the impact of circumstances

For the project activity, the baseline scenario identified at the validation of the project activity was the continuation of the current practice without any investment. The main investment environment or market characteristics especially the feed-in tariff, the policy in terms of market access permit have no significant change. The current practice for the baseline emissions is still the GHG emitted by NCPG: the equivalent electricity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources within the NCPG.

By verifying China Energy Statistical Yearbook 2013, China Energy Statistical Yearbook 2014, China Energy Statistical Yearbook 2015, China Energy Statistical Yearbook 2016, China Energy Statistical Yearbook 2017, China Energy Statistical Yearbook 2018, China Energy Statistical Yearbook 2019 and China Energy Statistical Yearbook 2020. It is confirmed that thermal power (which coal is the fuel) still domain the power supply in NCPG which is the same situation when project activity was under validation. Therefore, the main market characteristics have no change.

As the project is under normal operation which is in line with the original design thereby the condition used to determine baseline emissions in the previous crediting is still valid.

Step 1.3: Assess whether the continuation of use of current baseline equipment(s) or an investment is the most likely scenario for the crediting period for which renewal is requested

The current baseline scenario is the continuation of the current practice. In the absence of the project, the electricity would have been supplied by NCPG, and it will not request an investment by the project proponent or third party. So, this step is not applicable.

Step 1.4: Assessment of the validity of the data and parameters

The NCPG emission factor calculated ex-ante for the 1st crediting period needs to be updated, as per the “Tool to calculate the emission factor for an electricity system” version 07.0.

This parameter is properly described in the 3.3.6 of following report.

Conclusion on step 1:

CQC confirms that the current baseline is still valid as per methodology ACM0002 version 20.0. However, the grid emission factor needs to be updated for the subsequent crediting period.

Step 2: Update the current baseline and the data and parameters

As the baseline scenario of the project activity is still sustained in this crediting period, no update would be required. The NCPG emission factor will be updated as described in 3.3.6 of this report.

3.3.5 Additionality

Not applicable for the renewal of crediting period as no assessment would be required. The requirement of operation and construction of a wind power project is regulated in Renewable Energy Law of the People's Republic of China which remains no change comparing with the time of first validation. Therefore, it is able to confirm there is no surplus regulations required comparing with the first validation at time of renewal of crediting period.

3.3.6 Quantification of GHG Emission Reductions and Removals

The calculation of the emissions reductions exactly follows the procedures described in the methodology ACM0002 version 20.0 and relevant tool, e.g. the "Tool to calculate the emission factor for an electricity system" version 07.0.

CQC has assessed the calculation of project emissions, baseline emissions, leakage emissions and emission reductions. Corresponding calculations have been carried out based on calculation spreadsheet. The consistency of the parameters and equations presented in PD, as well as calculation spreadsheet etc., has been compared with the information and requirements presented in the methodology and respective tools.

The assumptions and data used to determine the emission reductions are listed in the PD and all the sources have been checked. Based on the information reviewed it is confirmed that the sources used are correctly quoted and interpreted in the PD. The values presented in the PD are considered reasonably based on the documentation and references reviewed and the results of the interviews.

The estimation of the emission reductions are considered correct as the calculations have been reproduced by the assessment team with the attainment of the same results.

CQC confirms methodology ACM0002 version 20.0 and relevant tool, e.g. the "Tool to calculate the emission factor for an electricity system" version 07.0 have been correctly applied to calculate baseline emissions, project emissions, leakage and net GHG emission reductions and removals.

Detailed information on the verification of the parameters used in the equations is found below. The algorithms for the determination of the baseline and project are discussed in the following sections.

The emission reductions are calculated by the difference between baseline emissions (BE_y), project emissions (PE_y) and leakage (L_y).

(1) Baseline emissions

As per the methodology ACM0002 version 20.0 that the baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The baseline emissions are calculated by multiplying the fixed baseline emission factor determined in the registered PDD of the project ex ante by net electricity supplied by the Project to the NCPG in the monitoring period.

Because the project activity is the installation of a new grid-connected renewable power plant/unit at a site where no renewable power plant was operated prior to the implementation of the project activity, the $EG_{PJ,y}$ equals to the Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh) ($EG_{facility,y}$). According to ACM0002, version 20.0, Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh) ($EG_{facility,y}$) is determined according to Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0.

The Combined margin CO₂ emission factor for grid connected power generation in year y ($E_{Fgrid,CM,y}$) is calculated in a transparent and conservative manner as a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the steps prescribed in the “Tool to calculate the emission factor for an electricity system”, version 07.0.

The PD version 02 completed on 29/08/2022 using the data for calculation of the grid emission factor at the time the PD was received for validation. The latest data available was from China Electric Power Yearbook 2016-2018, China Energy Statistical Yearbook 2016-2018. The calculation is in accordance with the calculation process of the combined margin emission factor published by the Chinese DNA: 2019 Baseline Emission Factors for Regional Power Grids in China published by the DNA of China on 29/12/2020.

The NCPG is selected as the electricity system of the project activity. Simple OM method (method a) is applied for calculating OM emission factor because according to the data from China Electric Power Yearbook 2014-2018, the low-cost/must-run resources in the latest five years (2013-2017) constitute less than 50% of the total grid generation. Ex-ante option with a 3-year generation weighted average is chosen to account for $EF_{grid,OMsimple,y}$. Option B of simple OM is selected for the calculation based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system (option A was denied because necessary data for Option A is not available). As a result, $EF_{grid,OMsimple,y}$ is calculated to be 0.9419 tCO₂e/MWh as a generation weighted average for the year 2015-2017.

Based on the most recent years energy balance of the NCPG, calculating the proportions of CO₂ emissions from the coal-fired, oil-fired and gas-fired power plants in total CO₂

emissions of thermal power plants and taking them as weight of each type of plant in the calculations;

Based on the most advanced commercialized technologies which are applied by the coal-fired, oil-fired and gas-fired power plants, calculating the emission factor of thermal power plants in NCPG. This approach is more conservative as it assumes all recently built plants have the fuel efficiency as that of the most advanced commercialized technologies;

Calculating the $EF_{grid,BM,y}$ through multiplying the emission factor of thermal power plants by the percentage share of thermal power plants installed capacity addition within all recently built installed capacity. The proper year is selected so that it is the closest time when the last 20% of installed capacity was built.

As a result, the $EF_{grid,BM,y}$ is calculated to be 0.4819 tCO₂e/MWh.

Based on the weight w_{OM} and w_{BM} of 0.75:0.25 by default for the second crediting period, the combined margin emission factor is calculated to be 0.8269 tCO₂e/MWh.

The annual net electricity supplied to the grid is 99,026MWh which is in line with the original design of the project. It's confirmed that the estimation of the figure is reasonable. Thereby, the baseline emissions could be calculated to be 99,026 MWh * 0.8269tCO₂e/MWh = 81,885 tCO₂e.

The assessment team confirmed that the emission reductions calculation is corrected in the PD. The values of the main parameters for calculating combined emission factor ($EF_{grid,CM,y}$) are crosschecked by the assessment team and the data sources are listed in below table:

Data and Parameters	Description	Data source
$EF_{grid,OM,y}$	Simple operating margin CO ₂ emission factor in year y	2019 Baseline Emission Factors for Regional Power Grids in China
$EF_{grid,BM,y}$	Build margin CO ₂ emission factor in year y	2019 Baseline Emission Factors for Regional Power Grids in China
w_{OM}	Weighting of operating margin emissions factor	Tool to calculate the emission factor for an electricity system, version 07.0
w_{BM}	Weighting of build margin emissions factor	Tool to calculate the emission factor for an electricity system, version 07.0
$EF_{grid,CM,y}$	North China Power Grid	Calculated based on above

	Baseline Emission Factor in 2019	parameters
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CQC confirms that all data sources and assumptions are appropriate, and calculations are correct, applicable to the proposed VCS project activity and will result in a conservative estimate of the emission reductions.

(2) Project emissions

According to ACM0002, version 20.0, for all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected.

Therefore, the project emission is considered as zero based on ACM0002, version 20.0. Therefore, $PE_y = 0 \text{ tCO}_2\text{e}$.

(3) Leakage

According to the methodology, the project activity leakage do not take into account, then no leakage according to ACM0002. Therefore, $L_y=0$.

(4) Emission reductions Based on the calculations and results presented in the sections above the implementation of the project activity will result in an average ex-ante estimation of emission reduction conservatively calculated to be 81,885tCO₂e per year for the selected 10 years crediting period. Total emission reductions during the second crediting period are estimated to be 818,850 tCO₂e.

3.3.7 Methodology Deviations

Not applicable.

3.3.8 Monitoring Plan

The project applies methodology ACM0002 version 20.0.

The electric energy metering system was equipped according to the requirements of the Technical administrative code of electric energy metering (DL/T448-2016).

Bidirectional meter M1 with accuracy class of 0.5S installed at the Tangtian Substation is used to measure the total electricity supplied to the grid by the Project and Phase II project ($EG_{\text{output},y}$) and the electricity imported from the grid by the project and Phase II project through the 110kV line ($EG_{\text{import},y}$). Grid company is responsible for the operation of M1, the meter M1 measured the electricity continuously, the readings are daily recorded by the project owner and the grid company, the monthly cut-off time is determined by the project owner and the grid company in advance every month. The reading records will be cross-checked with the sales receipts every month.

Meter M2 with accuracy class of 0.5S installed at the 110kV outline of the main transformer of the project on-site is used to measure the electricity supplied to the grid by the Project. The reading of meter M2 is carried out by the project owner for cross check.

Meter M3 with accuracy class of 0.5S installed at the 110kV outline of the main transformer of Phase II project on-site is used to measure the electricity generated by the Phase II project ($EG_{\text{another},y}$). The reading of meter M3 is recorded daily by the project owner and aggregated monthly for calculate the electricity supplied to the Grid by the project.

Additionally, a spare 10kV line is used for supplying electricity in order to start-up or emergencies. Electricity imported from the grid by the project via this spare 10kV line ($EG_{\text{mspare},y}$) is be measured by meter M4 with the accuracy of 1.0. The grid company is responsible for the operation of M4, the meter M4 is continuous measured and the readings are monthly recorded by the project owner and the grid company together on the appointed date. The reading records will be cross-checked with the sales receipts every month.

China Resources Wind Power (Qingdao) Co., Ltd., the project owner has used this document as guideline in monitoring of the project emission reduction performance and will adhere to the guidelines set out in this monitoring plan. This plan should be modified according to actual conditions and requirements of VVB in order to ensure that the monitoring is credible, transparent, and conservative.

An organizational structure is provided in PD. The functions such as data collection, aggregation, verification, calculation, archiving, as well as the maintenance of equipment etc. have been defined. Quality assurance and quality control procedures for recording, maintaining and data archiving etc. will be ensured according to VCS and CDM EB rules. The calibration of the meter will be implemented as per national standard. An emergency treatment process has been defined in PD when the meter is in malfunction. Data management and quality control system are quoted in PD. The monitoring staffs will be trained based on the training program described in PD.

CQC confirms that the monitoring plan contains all necessary parameters which have been clearly described in PD and that the means of monitoring described in the plan complies with the requirements of the methodology.

3.4 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VALIDATION CONCLUSION

CQC has performed a validation of renewal of crediting period of the “Qingdao Longxin Wind Power Project Phase I” (Ref. No. 969). The validation was performed on the basis of the updated sections of the PD relating to the baseline, estimated emission reductions and the monitoring plan using the most recent version of baseline and monitoring methodology applicable for the project activity. The final validation opinion was finalized in accordance with the VCS standard version 4.3, CDM VVS for project activities version 03.0 including the assessment of:

- (a) The impact of new relevant national and/or sectoral policies and circumstances on the baseline taking into account relevant guidance from the Board with regard to renewal of the crediting period of the registered VCS project activity at the time of requesting renewal of crediting period of the project activity;
- (b) The correctness of the application of the approved methodology and, where applicable, the approved standardized baseline for the determination of the continued validity of the baseline or its update, and the estimation of emission reductions for the applicable crediting period of the registered VCS project activity.

The review of the project design documentation and the subsequent follow-up interviews have provided CQC with sufficient evidence to determine the validity of the original baseline. The project correctly applies the latest baseline and monitoring methodology ACM0002, “Grid-connected electricity generation from renewable sources” version 20.0. CQC is able to confirm:

- (a) The updated PD complies with the valid version of the applicable PD form and instructions therein for filling out the PD;
- (b) Information transferred to the later valid version of the PD form is materially the same as that in the registered PD;
- (c) The baseline and monitoring methodology was applied in accordance with the applicable requirements in the VCS standard and Project Standard;
- (d) The baseline, the estimated GHG emission reductions, and the monitoring plan in the updated PD comply with the applicable requirements in the VCS standard and Project Standard, and the valid version of the methodology that is applicable to the registered VCS project activity;
- (e) The next crediting period of the registered VCS project activity commences on the day immediately after the expiration of the current crediting period;
- (f) The names of project participants in the updated PD are consistent with the names of the project participants in the registered PD.

Given that the project is implemented as designed and the underlying assumptions do not change, the project is likely to achieve the estimated amount of annual emission reductions of 81,885 tCO₂e and a total estimated emission reduction of 818,850 tCO₂e over the 2nd renewable crediting period as specified within the final PD.

In summary, it is CQC's opinion that the project activity "Qingdao Longxin Wind Power Project Phase I" (Ref. No. 969) in P. R. China, as described in the PD, version 02 dated 29/08/2022, meets all relevant VCS and UNFCCC requirements for the renewal of the crediting period.

APPENDIX 1: REFERENCE LIST

1. VCS PD, Version 01, dated 09/08/2022; version 02, dated 29/08/2022
2. ER calculation spreadsheet
3. CDM PDD, Version 03, dated 18/04/2012
4. CDM Validation Report, Version 01.2, dated 16/07/2012
5. TÜV Rheinland (China) Ltd. : VCS Validation Report for the project
6. VER/VCU Monitoring Report for the project from 08/11/2011- 27/09/2012
7. TÜV Rheinland (China) Ltd.: VCS Verification Report for the project from 08/11/2011- 27/09/2012
8. VCS standard version 4.3
9. VCS Program Guide, version 4.2
10. VCS Program Definitions, v4.2
11. VCS Validation Verification Manual v3.2
12. Approved methodology ACM0002, version 20.0
13. Tool to calculate the emission factor for an electricity system, version 07.0
14. Tool for the demonstration and assessment of additionality (version 07.0.0)
15. Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
16. Assessment of the validity of the original/current baseline and to update the baseline at the renewal of a crediting period, version 03.0.1
17. Tool to determine the remaining lifetime of equipment ,version 01
18. CDM Validation and Verification Standard for project activities version 03.0
19. CDM Project Standard for project activities version 03.0
20. CDM project cycle procedure for project activities version 03.0
21. Business license

22. Statement issued by the project proponent
23. Nameplate of the equipment
24. Feasibility Study Report of Qingdao Longxin Wind Power Project Phase I
25. China Energy Statistical Yearbook 2013-2020
26. 2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020
27. China Electric Power Yearbook 2016-2018
28. 2006 IPCC Guidelines for National Greenhouse Gas Inventories.